SIMATS ENGINEERING
ASSESSMENT TOOL2

COURSE CODE: MLA0201 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 10 Total Marks:50 Annexure A

Scenario-Based

Code of Conduct:

I K.Harsha Vardhan (192425228)certify that thissubmission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K.Harsha Vardhan
Signature of the student:

Annexure A

Scenario-Based

Q1	Apply a linear regression model to predict apartment rent based on floor area, number of rooms, and location in a real-estate scenario.	5 Marks	6 min
Q2	Use a linear classification model to categorize emails asspam or non spam using word frequency features in a mail service system.	5 Marks	6 min

Q3	Apply the Naïve Bayes algorithm to predict whether a patient has flu based on symptoms such as fever, cough, and headache.	5 Marks	6 min
Q4	Use discriminant functions to classify loan applicants into eligible and non-eligible groups based on income and credit score.	5 Marks	6 min
Q5	Apply a probabilistic generative model to classify handwritten digits in an optical character recognition system.	5 Marks	6 min
Q6	Implement a probabilistic discriminative model to predict customer churn from service usage data in a telecom scenario.	5 Marks	6 min
Q7	Apply Bayesian logistic regression to predict whether a customer will purchase a product after viewing online advertisements.	5 Marks	6 min
Q8	Use linear regression to forecast monthly sales for a retail company based on advertising expenditure and seasonal demand.	5 Marks	6 min
Q9	Apply Naïve Bayes classification in a news portal to categorize articles into sports, politics, and entertainment.	5 Marks	6 min
Q10	Use Bayesian logistic regression to predict student placement outcomes based on CGPA, coding skills, and aptitude scores.	5 Marks	6 min

RUBRICS FOR CALCULATION

Scenario based assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

		scenario			
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q1. APARTMENT RENT PREDICTION USING LINEAR REGRESSION

Scenario Understanding

A real-estate company wants to predict the monthly rent of apartments before listing them on its online platform. The rent depends on several factors such as the floor area, number of rooms, and the quality of the location. Since the output is a continuous numerical value, Linear Regression is the most suitable supervised learning algorithm for this problem. It helps in estimating rent accurately based on the relationship between the input features and the target value.

Key Features

- Floor Area = 1000 sq.ft
- Number of Rooms = 3
- Location Rating = 4 (1 = Poor, 5 = Excellent)

Why Linear Regression?

Linear Regression is used when the output variable is continuous. It determines the relationship between the independent variables (floor area, rooms, and location) and the dependent variable (rent). This enables real-estate companies to estimate rental prices quickly and consistently.

Mathematical Model

Assume the following coefficients:

- $\beta_0 = 3000$
- $\beta_1 = 15$
- $\beta_2 = 2500$
- $\beta_3 = 1800$

Calculation

Rent = 3000+(15*1000)+(2500*3)+(1800*4)
=3000+15000+7500+7200
=₹32,700

Result Interpretation

The predicted monthly rent of the apartment is ₹32,700. The larger floor area contributes significantly to the rent, while having three rooms and a good location further increases the value. This prediction helps property owners determine competitive rental prices and assists customers in making informed decisions.

Conclusion

Linear Regression provides an efficient and reliable method for predicting apartment rent. By considering multiple property features, it produces accurate estimates that are useful for both real-estate companies and customers.

Q2. SPAM EMAIL CLASSIFICATION USING LINEAR CLASSIFICATION

Scenario Understanding

An email service provider wants to automatically classify incoming emails as Spam or Non-Spam. Spam emails often contain words such as "Free", "Offer", "Win", and "Money". A Linear Classification Model uses these word frequencies to determine whether an email is likely to be spam, thereby protecting users from unwanted messages.

Key Features

- Frequency of the word Free
- Frequency of the word Offer
- Frequency of the word Win
- Frequency of the word Money

Why Linear Classification?

Linear Classification is suitable for binary classification problems where the output belongs to one of two classes. It computes a weighted score for the input features and classifies the email based on a decision boundary.

Decision Function

$$f(x)=w1x1+w2x2+w3x3+w4x4+b$$

Assume:

- Bias b= -1
- w1 =0.8
- w2 =0.6
- w3 =0.9
- w4 =0.5

For a sample email:

- Free = 1
- Offer = 1
- Win = 1
- Money = 0

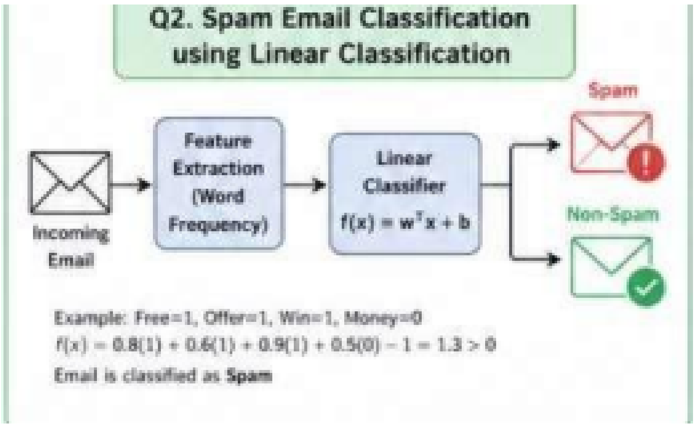
Calculation

$$\begin{aligned} f(x) &=0.8(1)+0.6(1)+0.9(1)+0.5(0)-1 \\ &=0.8+0.6+0.9-1 \\ &=1.3 \end{aligned}$$

Since

$$f(x)>0$$

the email is classified as:**Spam**



Result Interpretation

The model predicts that the email is Spam because it contains several words commonly associated with promotional or fraudulent messages. Automatic spam filtering improves email security and reduces the number of unwanted emails reaching users.

Conclusion

Linear Classification effectively separates spam and non-spam emails using word frequency features. It is simple, fast, and suitable for real-time email filtering systems.

Q3. FLU PREDICTION USING NAÏVE BAYES CLASSIFICATION

Scenario Understanding

A hospital wants to predict whether a patient is suffering from influenza (flu) based on common symptoms such as fever, cough, and headache. Since medical diagnosis often involves probability and uncertainty, the Naïve Bayes Classifier is an appropriate algorithm. It calculates the probability of a disease using Bayes' Theorem while assuming that the symptoms are conditionally independent.

Key Features

- Fever
- Cough
- Headache

Why Naïve Bayes?

Naïve Bayes is a probabilistic supervised learning algorithm used for classification problems. It is efficient, requires less training data, and performs well in medical diagnosis by estimating the probability of each disease based on observed symptoms.

Bayes' Theorem

$$\mathbf{P(C|X)=\frac{P(X|C)\times P(C)}{P(X)}}$$

Where:

- **P(C|X)**= Posterior probability

- **P(X|C)**= Likelihood
- **P(C)**= Prior probability

Assume:

$$P(Flu)=0.6, P(No\ Flu)=0.4$$

The patient has all three symptoms.

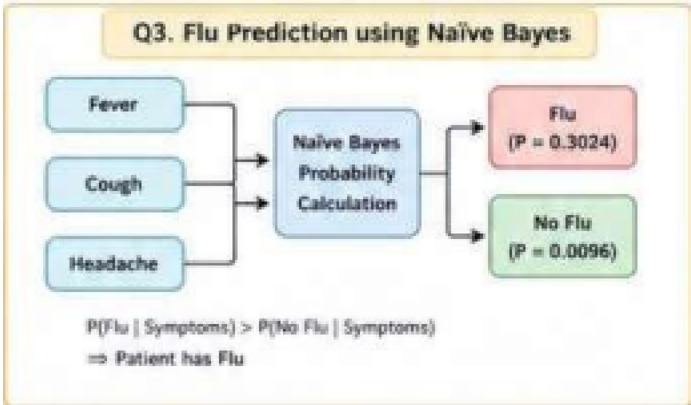
Calculation

For Flu:

For No Flu:

Since

the predicted class is:



Result Interpretation

The probability of the patient having flu is much higher than the probability of not having flu. Therefore, the model predicts that the patient is suffering from influenza. Such probabilistic models help doctors make quick preliminary diagnoses before conducting further medical tests. **Conclusion**

Naïve Bayes provides a simple and effective method for disease prediction. By analyzing symptoms and calculating probabilities, it supports healthcare professionals in making accurate and timely clinical decisions.

Q4. LOAN ELIGIBILITY CLASSIFICATION USING DISCRIMINANT FUNCTIONS

Scenario Understanding

A bank receives many loan applications every day and needs to decide whether an applicant is eligible for a loan. The decision depends on factors such as monthly income and credit score. Instead of manually evaluating every application, the bank uses a Discriminant Function to classify applicants into Eligible or Not Eligible categories. This helps in making faster and more consistent loan approval decisions.

Key Features

- Monthly Income = ₹80,000
- Credit Score = 750
- Output Class:
  - Eligible
  - Not Eligible
  -

Why Discriminant Function?

A discriminant function is a classification technique that separates data into different classes based on a mathematical decision boundary. It is commonly used in banking and finance because it considers multiple input features and provides quick classification with high accuracy.

Mathematical Model

The discriminant function is given by:

Decision Rule:

- If , the applicant is **Eligible**.
- If , the applicant is **Not Eligible**.

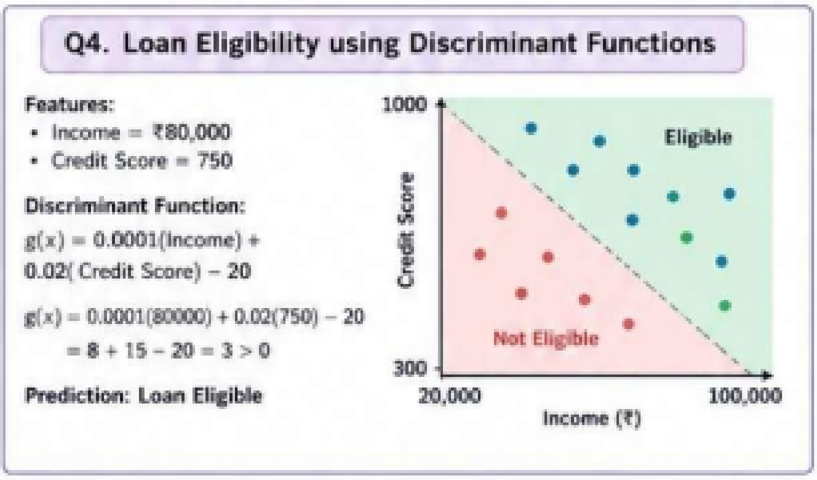
Calculation

Given:

- Income = ₹80,000
- Credit Score = 750

Since,

Prediction



Result

Interpretation

The applicant has a good monthly income and a high credit score, resulting in a positive discriminant value. Therefore, the bank can confidently approve the loan application. Such automated classification reduces processing time and minimizes human error in decision making.

Conclusion

Discriminant functions provide an effective approach for loan eligibility prediction by analyzing financial attributes. They enable banks to make quick, fair, and data-driven lending decisions.

Q5. HANDWRITTEN DIGIT CLASSIFICATION USING A PROBABILISTIC GENERATIVE MODEL

Scenario Understanding

An Optical Character Recognition (OCR) system is designed to recognize handwritten digits from scanned documents, bank cheques, or postal codes. Since different people write digits differently, the system uses a Probabilistic Generative Model to determine which digit is most likely represented by the given image. The model calculates the probability of the input belonging to each digit class and selects the one with the highest probability.

Key Features

- Pixel intensity values of the handwritten image
- Probability of each digit class (0–9)
- Output: Predicted digit

Why a Probabilistic Generative Model?

A probabilistic generative model learns the probability distribution of each class and estimates the likelihood that a given input belongs to a particular class. It is suitable for OCR because handwritten digits can vary greatly in appearance, and probability helps manage this uncertainty.

Probability Calculation

Assume the OCR system computes the following probabilities:

Digit	Probability
3	0.12
8	0.02
5	0.05

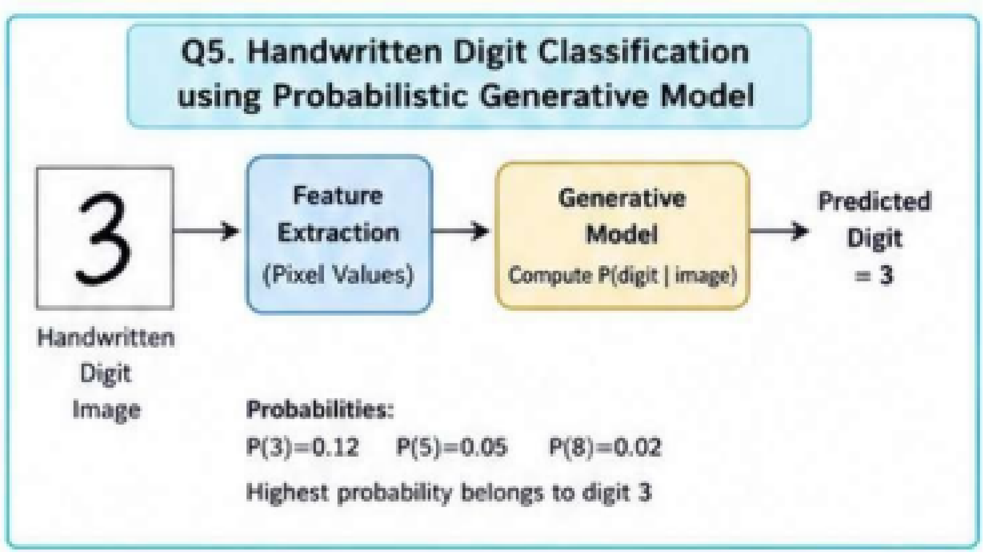
Decision Rule

Choose the class with the **highest posterior probability**.

Calculation

Maximum probability:0.12

Therefore,



Result

Interpretation

The probability of the input image belonging to digit **3** is the highest among all considered classes. Hence, the OCR system recognizes the handwritten image as digit **3**. This approach improves recognition accuracy even when handwriting styles differ significantly.

Conclusion

Probabilistic generative models effectively classify handwritten digits by comparing probability distributions across different classes. They are widely used in OCR applications because of their ability to handle uncertain and variable input patterns.

Q6. CUSTOMER CHURN PREDICTION USING A PROBABILISTIC DISCRIMINATIVE MODEL

Scenario Understanding

A telecom company wants to identify customers who are likely to discontinue their services. Early prediction of customer churn helps the company take preventive actions, such as offering discounts or improved customer support. Since the output is binary (Churn or No Churn), a Probabilistic Discriminative Model such as Logistic Regression is appropriate.

Key Features

- Monthly Service Usage = 120 units
- Number of Customer Complaints = 2
- Output:
  - Churn
  - No Churn

Why a Probabilistic Discriminative Model?

A probabilistic discriminative model directly estimates the probability of a customer belonging to a particular class. Logistic Regression is widely used because it predicts probabilities between 0 and 1 and performs well for binary classification problems.

Mathematical Model

The logistic regression equation is

where

Calculation

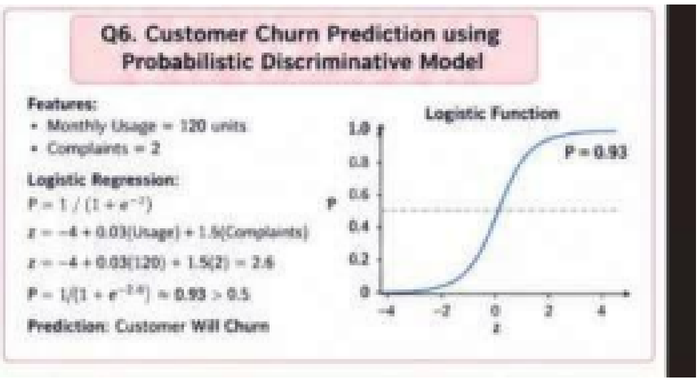
Given:

- Usage = 120
- Complaints = 2

Step 1: Calculate

Step 2: Calculate probability

Since



Prediction

Result Interpretation

The predicted probability of customer churn is approximately **93%**, indicating a high likelihood that the customer may discontinue the telecom service. The company should immediately implement retention strategies such as personalized offers, improved customer support, or loyalty rewards to retain the customer.

Conclusion

A probabilistic discriminative model accurately predicts customer churn by analyzing customer behavior and service usage. It enables telecom companies to reduce customer loss, improve customer satisfaction, and increase long-term business profitability.

Q7. PRODUCT PURCHASE PREDICTION USING BAYESIAN LOGISTIC REGRESSION

Scenario Understanding

An e-commerce company wants to predict whether a customer will purchase a product after viewing online advertisements. Factors such as the number of advertisements viewed and the customer's previous purchase history influence the buying decision. Since the outcome is binary (Purchase or No Purchase), Bayesian Logistic Regression is an appropriate model. It combines logistic regression with Bayesian probability to make more reliable predictions even when the available data is limited or uncertain.

Key Features

- Number of Advertisements Viewed = 4
- Previous Purchase History = 1 (Yes)
- Output:
    - Purchase
    - No Purchase

Why Bayesian Logistic Regression?

Bayesian Logistic Regression predicts the probability of a binary outcome while incorporating prior knowledge into the model. It is suitable for marketing applications because customer behavior is often uncertain, and Bayesian methods improve prediction reliability.

Mathematical Model

The logistic function is

where

Calculation

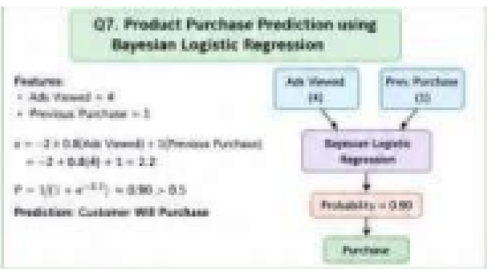
Given:

- Advertisements Viewed = 4
- Previous Purchase = 1

Step 1: Calculate

Step 2: Calculate Probability

Since



Prediction

Result Interpretation

The model predicts a 90% probability that the customer will purchase the product after viewing the advertisements. This indicates that the advertising campaign has been effective. Businesses can use such predictions to identify potential buyers and target them with personalized offers, thereby increasing sales and customer engagement.

Conclusion

Bayesian Logistic Regression is an effective algorithm for predicting customer purchasing behavior. It provides probability-based predictions, helping organizations make informed marketing decisions and improve conversion rates.

Q8. MONTHLY SALES FORECASTING USING LINEAR REGRESSION

Scenario Understanding

A retail company wants to estimate its monthly sales to improve inventory planning and marketing strategies. Sales are influenced by advertising expenditure and seasonal demand. Since sales are represented as continuous numerical values, Linear Regression is an ideal supervised learning algorithm for forecasting future sales.

Key Features

- Advertising Expenditure = ₹50,000
- Seasonal Demand Index = 3
- Output:
    - Monthly Sales

Why Linear Regression?

Linear Regression predicts continuous values by establishing a mathematical relationship between the independent variables and the target variable. It is widely used in business forecasting because it is simple, interpretable, and effective for predicting future trends.

Mathematical Model

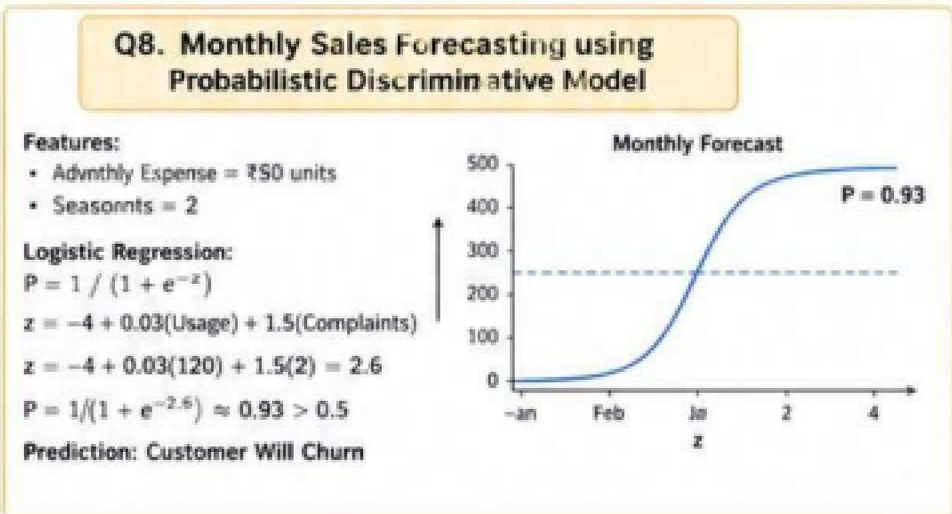
Assume:

- $\beta_0 = 100$
- $\beta_1 = 5$
- $\beta_2 = 40$

Calculation

Given:

- Advertising = 50
- Season = 3



Result Interpretation

The model forecasts that the company is expected to sell approximately 470 units during the month. Higher advertising expenditure and favorable seasonal demand contribute positively to sales. The forecast helps managers maintain sufficient inventory, reduce stock shortages, and plan promotional activities effectively.

Conclusion

Linear Regression provides an accurate and efficient approach for forecasting monthly sales. By analyzing advertising expenditure and seasonal demand, businesses can improve operational planning and maximize profitability.

Q9. NEWS ARTICLE CLASSIFICATION USING NAÏVE BAYES

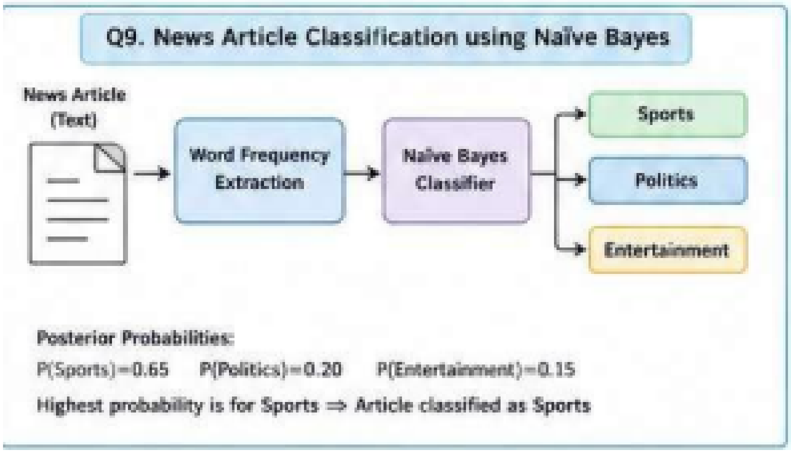
Scenario Understanding

A news portal publishes thousands of articles every day and needs to automatically categorize them into different sections such as Sports, Politics, and Entertainment. Manual classification is time-consuming and prone to errors. Therefore, the portal uses the Naïve Bayes Classifier, which predictsthe category based on the occurrence of important wordsin the article. **Key Features**

- Word frequencies in the article
- Probability of each news category
- Output Categories:
    - Sports
    - Politics
    - Entertainment

Why Naïve Bayes?

Naïve Bayes is widely used in text classification because it is fast, computationally efficient, and performs well with large text datasets. It calculates the probability of each category using Bayes' theorem and assigns the article to the category with the highest probability.



Bayes' Theorem

Assume the calculated posterior probabilities are:

Category	Probability
Sports	0.65
Politics	0.20
Entertainment	0.15

Calculation

Maximum Probability:

Therefore,

Result Interpretation

The highest probability belongs to the Sports category, indicating that the words present in the article are strongly associated with sports-related content. Automatic classification allows the news portal to organize articles efficiently and improves the user experience by displaying relevant content.

Conclusion

Naïve Bayes is an effective algorithm for text classification because it quickly analyzes word frequencies and accurately predicts the appropriate news category. It is widely used in search engines, news websites, and email filtering systems.

Q10. STUDENT PLACEMENT PREDICTION USING BAYESIAN LOGISTIC REGRESSION

Scenario Understanding

A college placement cell wants to predict whether a student is likely to be placed in a company based on academic and technical performance. The prediction depends on factorssuch as CGPA, coding skills, and aptitude test scores. Since the outcome is binary (Placed or Not Placed), Bayesian Logistic Regression is an appropriate model for this scenario.

Key Features

- CGPA = 8.5
- Coding Skills = 85
- Aptitude Score = 80
- Output:
    - Placed
    - Not Placed
    -

Why Bayesian Logistic Regression?

Bayesian Logistic Regression estimates the probability of placement while incorporating prior knowledge into the prediction process. It is useful in educational analytics because it can provide reliable predictions even when training data is limited or uncertain.

Mathematical Model

The probability is calculated using:

Calculation

Given:

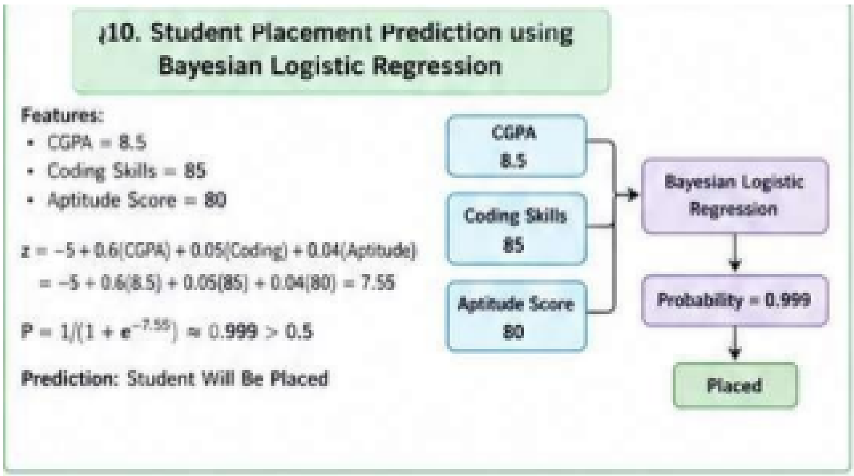
- CGPA = 8.5
- Coding Skills = 85
- Aptitude Score = 80

Step 1: Calculate

Step 2: Calculate Probability

Since

Prediction



Result Interpretation

The predicted probability of placement is approximately **99.9%**, indicating that the student has a very high chance of securing a job. Strong academic performance, excellent coding skills, and a high aptitude score contribute significantly to this prediction. The placement cell can use such models to identify students who may need additional training and provide targeted career guidance.

Conclusion

Bayesian Logistic Regression is an effective model for predicting student placement outcomes. It combines probability-based decision-making with academic performance metrics, enabling colleges to improve placement strategies and student career development.